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Circulating tumor DNA level is associated with time to clinical recurrence in Merkel cell carcinoma: Implications for patient management

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Abstract

Background: Merkel cell carcinoma (MCC) recurs in 40% of patients. Circulating tumor DNA (ctDNA) is an emerging blood-based biomarker for early MCC recurrence detection.

Objective: To evaluate the timing and prognostic significance of ctDNA levels relative to clinical recurrence.

Methods: This multicenter prospective study analyzed 669 tumor-informed ctDNA tests from 215 MCC patients (stage I-IV) without clinically evident disease after treatment.

Results: Patients with at least one positive ctDNA test were more likely to experience recurrence compared to ctDNA-negative patients (hazard ratio: 18.1, 95% CI: 8.9–36.7), with 77% developing clinically evident disease by one year. The median interval between the first positive ctDNA and clinical recurrence was 2.7 months. Clinical recurrences usually occurred within 3 months for ctDNA levels above 10 molecules/mL, within 6 months for levels between 1–10 molecules/mL, and within 9 months for levels below 1 molecule/mL.

Limitations: In this real-world study, there was variability in timing and frequency of follow-up examinations, imaging, and ctDNA testing, although most patients were followed with both ctDNA and imaging.

Conclusions: A positive ctDNA test detects MCC recurrence approximately 3 months earlier than imaging. Negative ctDNA can help reduce imaging frequency through serial ctDNA monitoring, while positive ctDNA warrants closer patient follow-up.

Capsule summary

- Tumor-informed circulating tumor DNA (ctDNA) is extremely sensitive for detecting Merkel cell carcinoma recurrence, although data-driven guidance of its clinical integration is limited.
- ctDNA-negative patients require less frequent imaging. For ctDNA-positive patients without clinically evident disease, the specific ctDNA level can guide imaging frequency, enabling more personalized surveillance strategies.

Keywords

Merkel cell carcinoma; circulating tumor DNA; ctDNA; biomarker; recurrence; lead time

Introduction

Merkel cell carcinoma (MCC) is a rare cutaneous neuroendocrine cancer, with approximately 80% of cases being Merkel cell polyomavirus (MCPyV)-positive and 20% virus-negative in the United States. MCC is aggressive, with a recurrence rate of 40%.¹ Circulating tumor DNA (ctDNA) has emerged as a blood-based tumor biomarker for MCC recurrence surveillance, regardless of virus status. We recently reported that tumor-informed ctDNA has high sensitivity and specificity for detecting clinically evident disease.² Furthermore, in patients with locoregional (stage I–III) MCC, a positive ctDNA result within four months following curative treatment was strongly associated with an elevated risk of recurrence (hazard ratio [HR], 7.4; $p < 0.001$) in patients without clinically detectable disease.² These findings led to the inclusion of ctDNA in the 2025 NCCN guidelines for MCC surveillance.³

Despite this progress, there remains a lack of published evidence regarding how to effectively integrate ctDNA into routine clinical practice, particularly if ctDNA is positive in the absence of clinically evident disease.⁴ ctDNA positivity often precedes clinical recurrence; however, data on lead times between initial ctDNA detection and clinical recurrence have not been reported.⁴

To address this unmet need, we conducted secondary analyses of a previously published dataset, focusing on the temporal relationship between the first positive ctDNA result and subsequent clinical recurrence. This study aims to provide practical, data-driven guidance to support surveillance planning and enhance clinical decision-making in patients with MCC undergoing ctDNA monitoring.

Materials and methods

Study Design and Patient Cohort

This study was designed as a prospective, longitudinal, and observational study from six MCC tertiary centers in the United States, including Stanford University, University of Washington, Dana-Farber Cancer Institute, Northwestern University, University of California San Francisco, and Moffitt Cancer Center.² The research was approved by the Institutional Review Board (IRB) at each participating institution. All patients provided written informed consent. Additionally, a data-sharing agreement was established between the centers involved. Patients enrolled from April 2020 to August 2022, with a data cutoff date of July 8, 2022, for Stanford University and University of Washington, and August 31, 2022 for the other four sites. Patients were eligible to enroll in the primary study at any point during their disease course, including before initiating treatment or after having received treatment. All patients were staged according to the American Joint Cancer Committee (AJCC) eighth edition staging system.⁵ To determine Merkel cell polyomavirus status, the MCPyV oncoprotein antibody serology assay was utilized.⁶

Stage I–IV patients who underwent ctDNA testing during surveillance after confirming no clinical evidence of disease were included in this secondary analysis. Blood was collected for ctDNA testing at enrollment and approximately every 3 months during surveillance.

Imaging with computed tomography (CT), positron emission tomography (PET)-CT, magnetic resonance imaging, or ultrasound, was performed at enrollment. Unexpected rises in ctDNA were followed by an additional ctDNA test coupled with an imaging study within 4 weeks (if >6 weeks since last imaging). Clinically evident disease was defined as MCC detected through physical examinations or imaging, followed by tissue biopsy whenever feasible. To capture ctDNA that was performed more or less concurrently with an imaging test, any ctDNA tests performed within 14 days after recurrence were treated as if conducted on the recurrence date. Patients were excluded if they had insufficient clinical data, or if ctDNA assay development was impossible due to insufficient tumor tissue or prior hematopoietic stem cell transplantation.

Tumor-informed circulating tumor DNA assay

A tumor-informed ctDNA assay (Signatera™, Natera Inc, Austin, TX, USA) was utilized as previously described.⁷ Tumor tissue and matched normal blood from each patient underwent whole-exome sequencing to identify up to 16 patient-specific somatic single nucleotide variants (SNVs). Subsequently, personalized multiplex PCR-NGS assays were designed to detect ctDNA in the corresponding patients' plasma samples. Samples were considered positive if two or more somatic SNVs were identified. ctDNA concentrations were reported as mean tumor molecules (MTM) per mL of plasma, with values greater than 0.00 indicating positivity and values equal to 0.00 classified as negative.

Outcomes

The ctDNA surveillance period was defined as starting at the first ctDNA test performed after a patient was determined to be clinically negative for MCC until clinical recurrence or end of follow-up. Determination of clinical recurrence was based on findings from physical examinations and imaging studies and was confirmed by biopsy whenever feasible. The date of clinical recurrence was defined as the date of biopsy confirming recurrence or the date of imaging that first identified the recurrence, whichever occurred earlier. Death was considered a competing risk and time to clinical recurrence was censored at the last known follow-up or the predefined study cut-off date.

Statistical Analysis

Statistical analyses were conducted using R version 4.0.3 (R Foundation for Statistical Computing, Vienna Austria). All p-values were two-sided, with statistical significance defined as $p < 0.05$. Comparisons between groups were conducted using Fisher's exact test or the Wilcoxon rank-sum test.

The association between longitudinal ctDNA status (anytime positive vs. persistently negative) and the risk of clinical recurrence during ctDNA surveillance was assessed using Cox proportional hazards regression, with ctDNA status treated as a time-varying covariate to avoid immortal time bias.⁸ The time-varying ctDNA status was defined as negative if the current and prior ctDNA tests were all negative and as positive if the current or any prior ctDNA test was positive. Other factors were included in the models: sex, age, immunosuppression status, baseline MCPyV oncoprotein antibody status, history of recurrence, and history of regional or distant disease.

Among patients who became ctDNA positive at any point during surveillance, the median time to recurrence was estimated as the time from the first positive ctDNA test to when the estimated risk of recurrence first reached 50% or greater. Lead time was calculated as the time from the first positive ctDNA test to clinical recurrence using the 29 patients who had been ctDNA positive within 6 months of developing their recurrence. Additional details are included in the supplemental methods.

Results

A total of 215 patients underwent ctDNA testing during surveillance, with a median follow-up of 233 days. Table 1 summarizes the baseline clinical characteristics assessed at the start of ctDNA surveillance. In total, 669 ctDNA tests were performed across all patients, with a median between-test interval of 87 days (inter-quartile range [IQR]: 63–102).

Among the 215 patients, 33 were ctDNA-positive at their first test during surveillance, despite no clinical evidence of disease at that time, and 182 were ctDNA-negative (0 MTM/mL). Of the 182 initially ctDNA-negative patients, 31 patients converted from ctDNA-negative to ctDNA-positive during surveillance (median time to positive ctDNA: 4.6 months, range: 1.1–20.5 months), for a total of 64 patients with at least one positive ctDNA test during surveillance. Among the patients who were initially ctDNA-negative and remained ctDNA-negative, the overall risk of clinical recurrence during the first 3, 6, and 12 months of surveillance was 0.6% (95% CI: 0.1–4.0%), 4.6% (95% CI: 2.2–9.5%), and 7.7% (95% CI: 4.1–15%), respectively. ctDNA positivity at any point during surveillance was strongly associated with an increased risk of subsequent clinical recurrence, both before (hazard ratio [HR]: 18.8, 95% CI: 10.1–40.8) and after adjustment for other risk factors (adjusted HR [aHR]: 18.1, 95% CI: 8.9–36.7; Figure 1).

Among the 64 ctDNA-positive patients, the overall risk of clinical recurrence within 3-, 6-, and 12-months following the first positive ctDNA test was 43% (95% CI: 29–55%), 52% (95% CI: 36–63%), and 77% (95% CI: 56–88%), respectively (median time to recurrence: 5.7 months; Figure 2). The ctDNA level of the first positive ctDNA test ranged from 0.04 to 1883 MTM/mL (median: 2.0 MTM/mL, IQR: 0.6–5.4 MTM/mL). The risk of recurrence, stratified by ctDNA levels into three categories (<1 MTM/mL, 1–10 MTM/mL, and >10 MTM/mL), is shown in Figure 2, with corresponding median times to recurrence of 7.5 months, 2.8 months, and 1.3 months, respectively. Notably, while higher ctDNA levels were associated with a shorter time to clinical recurrence, the estimated risk of recurrence exceeded 70% by 12 months (or at the end of follow-up) across all three ctDNA level categories.

In univariable analyses, a higher ctDNA level at the first positive test was strongly associated with subsequent rate of clinical recurrence ($p < 0.001$) while no other risk factors had a statistically significant association, including baseline MCPyV oncoprotein antibody status ($p = 0.15$), history of recurrence ($p = 0.65$) and history of regional or distant disease ($p = 0.83$; Table 2). After adjusting for other risk factors, the ctDNA level at the first positive test remained significantly associated with time to clinical recurrence (aHR: 2.5 per 10-fold increase in ctDNA; 95% CI: 1.5–3.9).

A total of 50 patients experienced clinical recurrences, with molecular evidence of disease detected via positive ctDNA either concurrent with recurrence (n=9) or preceding it (n=29) in 38 patients (76%). Among these 38 recurrences, distant recurrences were the most common (n = 25, 66%), followed by regional/in-transit (n = 11, 29%) and local (n = 2, 5%). The ctDNA level at the time of clinical recurrence ranged from 0.2 to 4625 MTM/mL (median: 20.5 MTM/mL, IQR: 4.5–202.6 MTM/mL). There was a median 19-fold increase in ctDNA level ($p < 0.001$) between the first positive test and when disease became clinically evident among the 23 patients who had at least two positive ctDNA tests by the time of the recurrence. The median ctDNA level of the initial positive test was numerically higher for distant recurrences compared to locoregional recurrences, though the difference was not statistically significant (median: 3.7 [IQR: 2.0–9.6] vs. 1.9 [IQR: 0.6–5.4] MTM/mL, $p = 0.20$).

For the 29 recurrences preceded by ctDNA positivity, the median lead time from the first positive ctDNA to clinical recurrence was 2.7 months (95% CI: 1.9–4.7, IQR: 1.8–5.7, range: 0.8–12.9; Figure 3).

Three patients did not have ctDNA testing around the time they developed a clinical recurrence, making these cases non-evaluable for further analysis. There were 9 patients who did not have a positive ctDNA test by the time of their clinical recurrence. Of these, 5 became ctDNA-positive shortly after their clinical recurrence (median: 22 days; distant n=2, regional n=3). The other 4 remained ctDNA-negative throughout follow-up despite developing locoregional disease (regional n=2, in-transit n=2; Figure 3). Notably, 2 of these 4 patients had small, in-transit superficial cutaneous recurrences near the primary MCC site, which were only detected through physical examinations and biopsy. Imaging was negative for both recurrences. Additionally, one of the remaining ctDNA-negative patients had suspected clinical recurrence based on low-level fluorodeoxyglucose (FDG) uptake in non-enlarged axillary lymph nodes and was started on immunotherapy; however, this suspected recurrence was not confirmed pathologically. The last patient was considered to have a second primary MCC based on anatomic location, which was later confirmed through biopsy and molecular studies. Of note, ctDNA based on the sequence of one primary tumor would not be expected to identify the presence of a second, unrelated primary tumor.

Discussion

The clinical utility of ctDNA for MCC recurrence detection is actively being explored. A recent study showed that ctDNA has high sensitivity and specificity for clinically evident MCC as well as the ability to detect subclinical disease that persists following initial surgery and or radiation therapy.^{2,9–15} However, a recent consensus meeting highlighted the need for data regarding how this emerging technology can best be integrated with imaging and with routine patient care.⁴

The present study evaluated the interval between an initial positive ctDNA test and clinical recurrence. An increase in ctDNA preceded clinical recurrence with a median lead time of 2.7 months. Overall, 77% of patients with positive ctDNA developed clinically evident disease by one year. More specifically, in most patients, clinical recurrence arose within 3

months if ctDNA was above 10 MTM/mL, 6 months if ctDNA was 1–10, and 9 months if ctDNA was below 1. Notably, patients with low ctDNA (<1) demonstrated delayed recurrence, with only 23% recurring within three months.

Traditionally, physicians rely on routine physical examinations and imaging every three months during the first two years for MCC surveillance, especially in higher-risk patients. However, the advent of ctDNA has the potential to reshape surveillance paradigms, with patients able to be monitored using ctDNA every three months during this time period. This study provides actionable insights into how ctDNA can inform MCC surveillance strategies. Lower-risk patients with consistently negative ctDNA results may benefit from an imaging de-escalation approach such as extending the interval between routine scans that reduces unnecessary radiation exposure and financial burden. This approach aligns with our prior data showing that ctDNA negativity predicted outcomes over the following 135 days with a 93% negative predictive value (95% CI: 89–97%).² Our data suggest that ctDNA, when combined with a physical examination, effectively detects recurrences and does not overlook cases that would have otherwise been identified through imaging. Conversely, a positive or rising ctDNA level warrants prompt imaging to detect potential clinically evident disease. If imaging is negative at the time of initial ctDNA positivity, frequent imaging may not be required, as some patients do not experience clinical recurrence for many months to even more than a year. Thus, surveillance strategies should be tailored to a patient's ctDNA level and recurrence risk.

While the role of ctDNA in MCC recurrence surveillance is increasingly established, how it can best be integrated into MCC treatment management remains less clear. Specifically, for patients with ctDNA positivity but no clinical evidence of disease, there is no consensus as to whether systemic immunotherapies should be initiated. Prospective studies, such as randomized clinical trials, are needed to address this gap.

This study has several limitations. Due to its real-world nature, there are variations in treatment protocols and follow-up intervals, including the timing of physical examinations, imaging, and the frequency of ctDNA testing across the six centers. These variations in timing can affect the observed lead times. Furthermore, the cohort was heterogenous in terms of stage, type and timing of prior treatment, and history of recurrence. While factors such as stage and history of recurrence were adjusted for in the analysis, the type and timing of prior treatments could not be accounted for. As a result, the findings may not fully reflect the outcomes of patients who enter surveillance immediately following definitive treatment. Lastly, a positive ctDNA test might have led to intensified surveillance, including more frequent imaging or physical exams, which could have reduced the apparent lead time observed in this study compared to a cohort with exams strictly performed at standard 3-month intervals. Despite these limitations, this study captured ctDNA dynamics within the context of routine clinical care. This analysis provides pragmatic guidance relevant to surveillance practice for MCC.

Conclusion

In summary, ctDNA can detect recurrent disease earlier than traditional imaging and physical examinations. Understanding the lead time between the first ctDNA positivity and clinical recurrence can help tailor follow-up frequency and urgency, optimizing patient-specific care. This approach offers an opportunity to de-escalate imaging frequency for ctDNA-negative patients, reducing unnecessary interventions while maintaining vigilance in higher-risk individuals.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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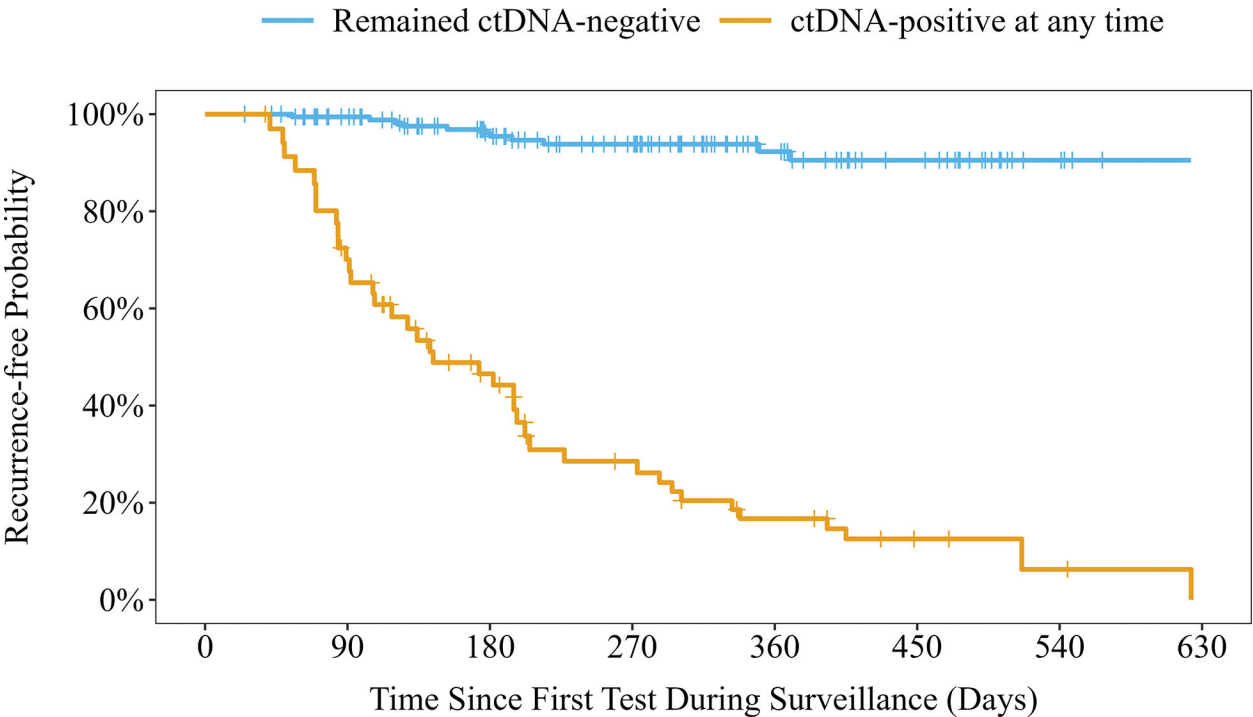
Abbreviations

MCC	Merkel cell carcinoma
MCPyV	Merkel cell polyomavirus
ctDNA	Circulating tumor DNA
HR	Hazard ratio
NCCN	National Comprehensive Cancer Network
IRB	Institutional Review Board
AJCC	American Joint Cancer Committee
CT	Computed tomography
PET	Positron emission tomography
SNVs	Single nucleotide variants
MTM	Mean tumor molecules
IQR	Interquartile range
aHR	Adjusted hazard ratio

CI Confidence interval

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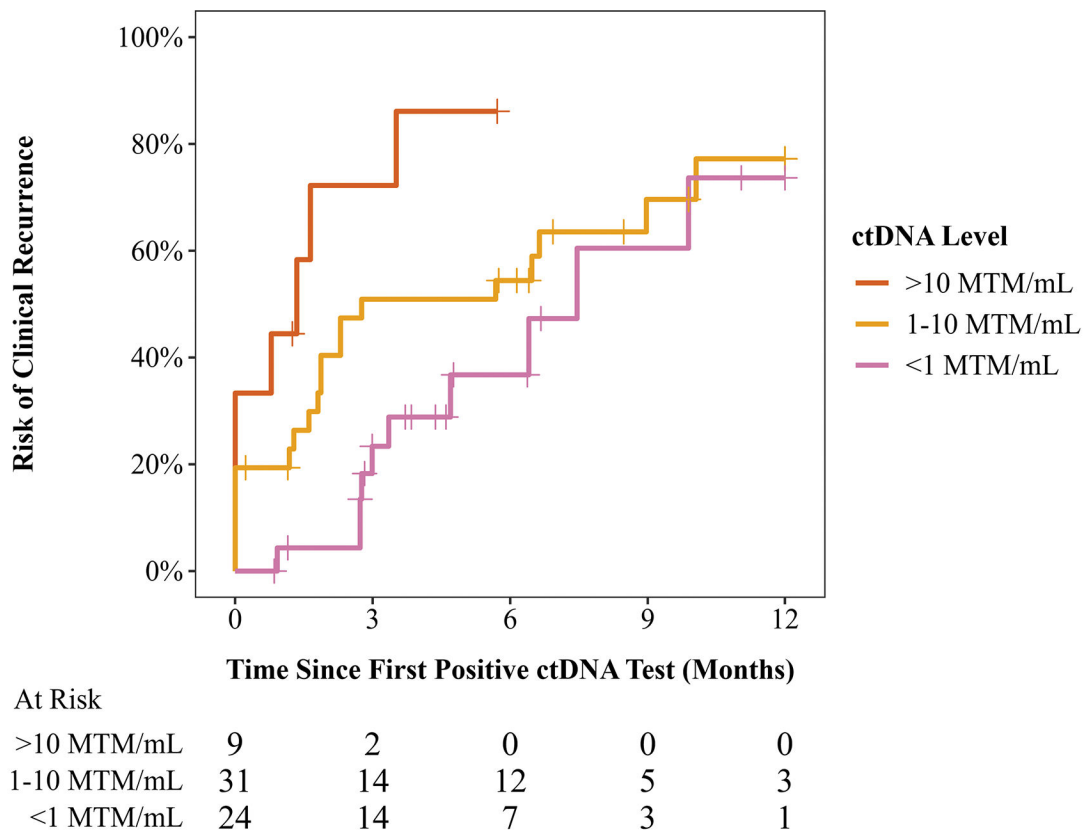


At risk

ctDNA(-)	182	155	119	81	46	22	5	0
ctDNA(+)	33	29	19	12	9	3	1	0

Figure 1. Proportion of patients who remained free of clinical recurrence during surveillance of their Merkel cell carcinoma, stratified by circulating tumor DNA status

Risk of recurrence by ctDNA status during the surveillance period as determined by a Cox regression model with ctDNA status as a time-varying covariate and adjusting for other risk factors. The recurrence-free probability after a positive ctDNA test (orange) at any point during disease course was significantly lower than when ctDNA tests were persistently negative (blue) (aHR: 18.1, 95% CI: 8.9–36.7). The recurrence-free probability at 1 year was 92% (95% CI: 85–96%) in the ctDNA-negative group and 17% (95% CI: 3–28%) in the ctDNA-positive group.



Estimated risk of recurrence after the first positive ctDNA test at each time point

Group	No.	3 Months		6 Months		9 Months		12 Months		Median Months to Recurrence
		Risk	(95% CI)	Risk	(95% CI)	Risk	(95% CI)	Risk	(95% CI)	
Overall	64	43%	(29-55%)	52%	(36-63%)	68%	(49-80%)	77%	(56-88%)	5.7
By ctDNA level										
>10 MTM/mL	9	72%	(13-91%)	86%	(16-98%)	*		*		1.3
1-10 MTM/mL	31	51%	(29-66%)	54%	(32-69%)	70%	(44-84%)	77%	(47-90%)	2.8
<1 MTM/mL	24	23%	(3-39%)	37%	(9-56%)	60%	(15-82%)	74%	(21-91%)	7.5

Figure 2.

Risk of clinical recurrence of Merkel cell carcinoma after a first positive ctDNA test, stratified by ctDNA level

*The risk of recurrence by 9, and 12 months could not be estimated in this group because all 9 patients had either been diagnosed with recurrence (n = 7) or were censored at the data cutoff (n = 2) by that time.

aHR = adjusted hazard ratio; CI = confidence interval; MTM = mean tumor molecule

A strong association was observed between ctDNA level and clinical recurrence risk (aHR: 2.5 per 10-fold increase in ctDNA, 95% CI: 1.5–3.9).

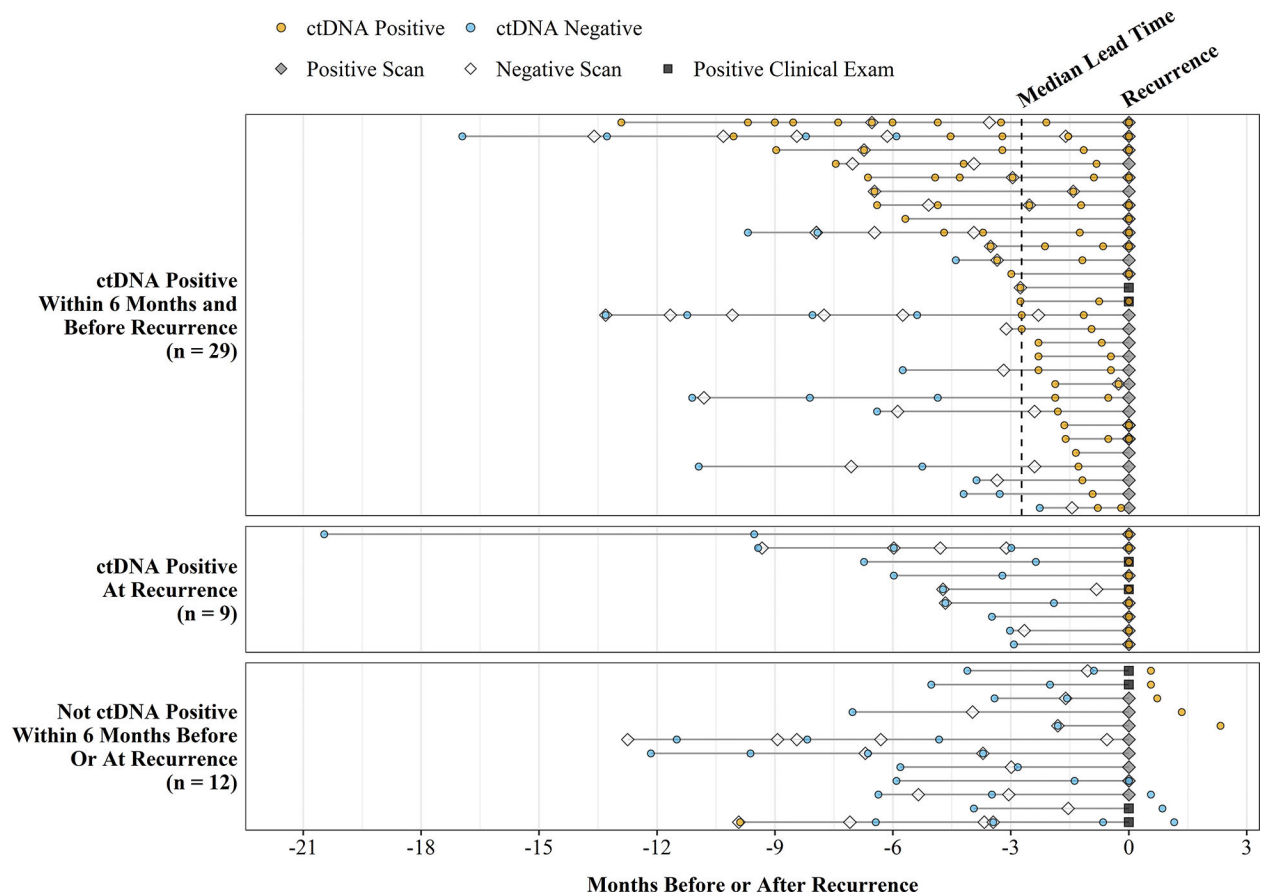


Figure 3.

Swimmer plot depicting each patient's timeline up to clinical recurrence of Merkel cell carcinoma

The longitudinal timeline for each patient's ctDNA assessments (circles) and imaging evaluations (diamonds) is depicted, starting from the first surveillance ctDNA test or scan (left) and progressing to confirmed clinical recurrence (right). Instances where clinical recurrence was initially identified through physical examinations rather than imaging are marked with a dark square at the recurrence time point. Of the 50 recurrences, ctDNA was positive either within 6 months before recurrence (n = 29) or at the time of recurrence (n = 9), accounting for 38 cases (76%). Among the 29 cases with ctDNA positivity prior to recurrence, the median lead time from the first positive ctDNA test to recurrence was 2.7 months (95% CI: 1.9–4.7; IQR: 1.8–5.7; range: 0.8–12.9). When stratified by initial ctDNA levels (<1 MTM/mL, 1–10 MTM/mL, and >10 MTM/mL), the median lead times were 3.0 months (n = 9), 2.3 months (n = 16), and 1.5 months (n = 4), respectively. For patients who were not ctDNA-positive at the time of recurrence, ctDNA results from follow-up assessments conducted within 90 days of clinical recurrence are included when available. Patients who were not ctDNA-positive by the time of their recurrence are described in more details in the results section.

Table 1.
Patient characteristics at start of circulating tumor DNA surveillance in Merkel cell carcinoma

Variable	All Patients (n = 215)	ctDNA-Positive Patients* (n = 64)
Male sex	142 (66%)	48 (75%)
Age, years	71 (41 – 97)	76 (57 – 97)
Immunosuppressed	37 (17%)	14 (22%)
MCPyV oncoprotein antibody status at MCC diagnosis		
Positive	111 (53%)	32 (51%)
Negative	98 (47%)	31 (49%)
Unavailable	6	1
History of recurrence	52 (24%)	17 (27%)
Most advanced stage in disease course		
Local	73 (34%)	19 (30%)
Regional	113 (53%)	37 (58%)
Distant	29 (13%)	8 (12%)
AJCC 8 th ed. stage at MCC diagnosis		
pI	43 (20%)	9 (14%)
cI	18 (8%)	7 (11%)
pII	16 (7%)	5 (8%)
cII	15 (7%)	4 (6%)
pIIIA	68 (32%)	21 (33%)
pIIIB	44 (20%)	15 (24%)
cIII	5 (2%)	2 (3%)
IV	6 (3%)	1 (2%)

Values are no. (%) or median (range); MCPyV = Merkel cell polyomavirus; AJCC = American Joint Committee on Cancer.

* Patients who were ctDNA-positive at any point during their surveillance period.

Table 2.

Risk of subsequent Merkel cell carcinoma clinical recurrence relative to ctDNA level at the time of first positive test and other clinical risk factors

Variable	Univariable Models			Multivariable Model		
	HR	(95% CI)	P-value	aHR	(95% CI)	P-value
Higher ctDNA level [*]	2.4	(1.6–3.6)	<0.001	2.5	(1.5–3.9)	<0.001
Male sex	0.9	(0.4–1.9)	0.80	1.3	(0.6–3.2)	0.50
Immunosuppressed	1.6	(0.8–3.3)	0.19	1.2	(0.5–2.7)	0.72
Older Age [†]	1.1	(0.7–1.6)	0.77	0.9	(0.6–1.5)	0.74
History of regional or distant disease	0.9	(0.4–1.9)	0.83	0.8	(0.3–2.0)	0.70
History of recurrence	0.8	(0.4–1.7)	0.65	0.7	(0.3–1.7)	0.46
Positive baseline MCPyV oncoprotein antibody status [‡]	0.6	(0.3–1.2)	0.15	0.7	(0.3–1.5)	0.34

HR = hazard ratio; aHR = adjusted hazard ratio; CI = confidence interval; MCPyV = Merkel cell polyomavirus

^{*} Per 10-fold increase for ctDNA level at first positive test

[†] Per 10-year increase of age, at time of first positive ctDNA test

[‡] Baseline MCPyV oncoprotein antibody status was not available for one patient, who was excluded from the corresponding univariable and multivariable models.